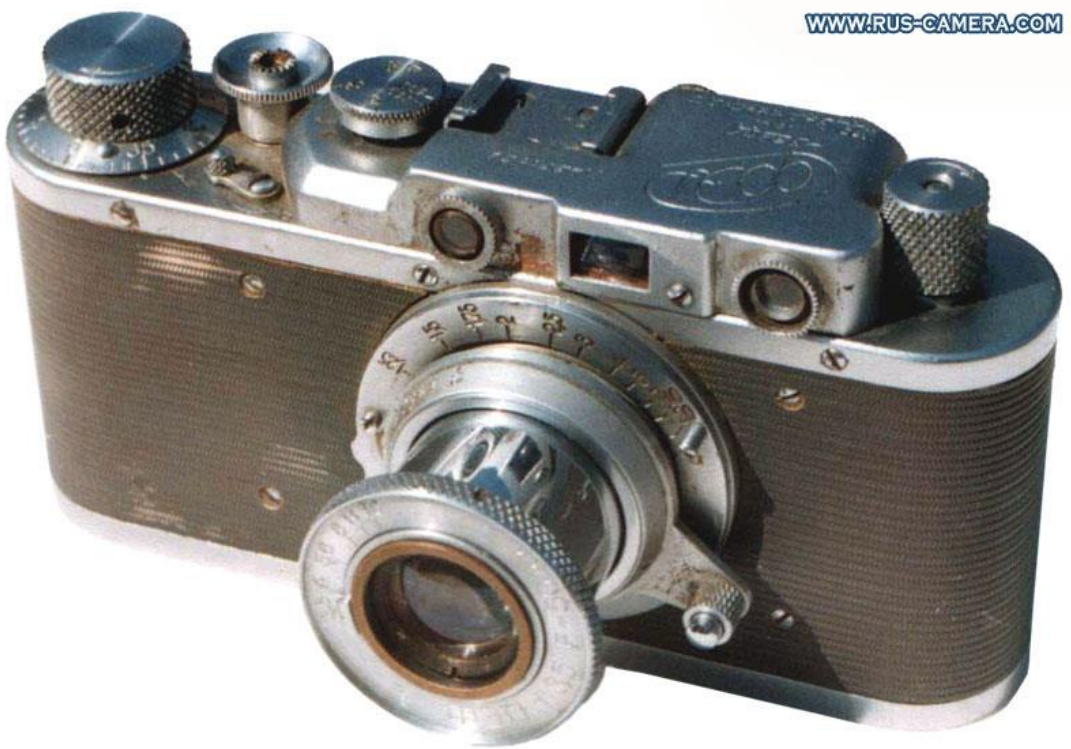
# CMAI 480

# Computer Vision

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Computer Science and Engineering Department
Qatar University

# Cameras



# Outline

- Cameras Geometry and image formation

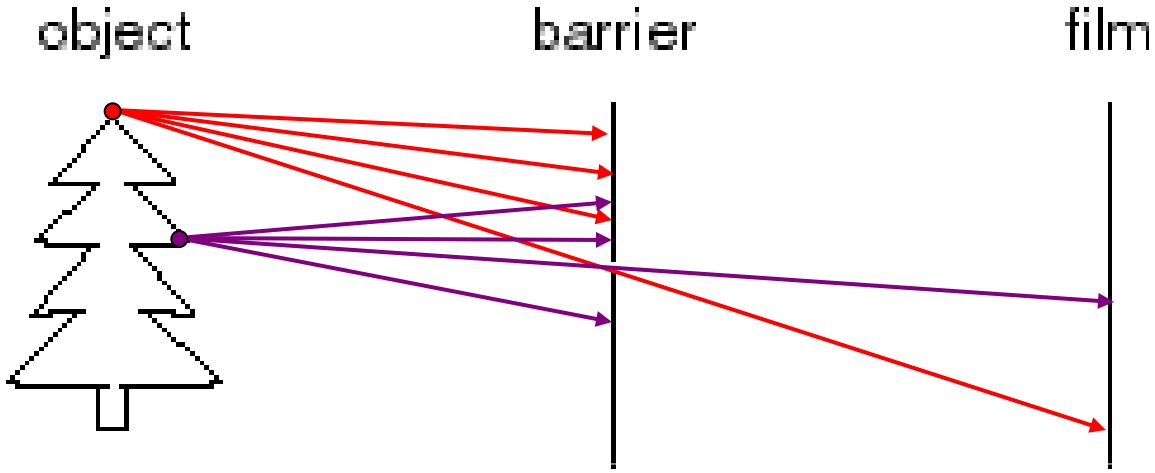
# Cameras

- Recall:
  - A camera is a mapping from the 3D world to a 2D image.
  - Our primary focus will be on cameras that carry out Central Projection.
    - Central projection means that all rays that connect a world point with its corresponding image point pass through a common point known as the Center of Projection (CoP).

# Camera Models

- At this time we are interested in two special cases of the central projection cameras:
  1. When the COP is at a finite location; and
  2. When the COP is at infinity
- A good place to start for understanding both is the basic pinhole camera.

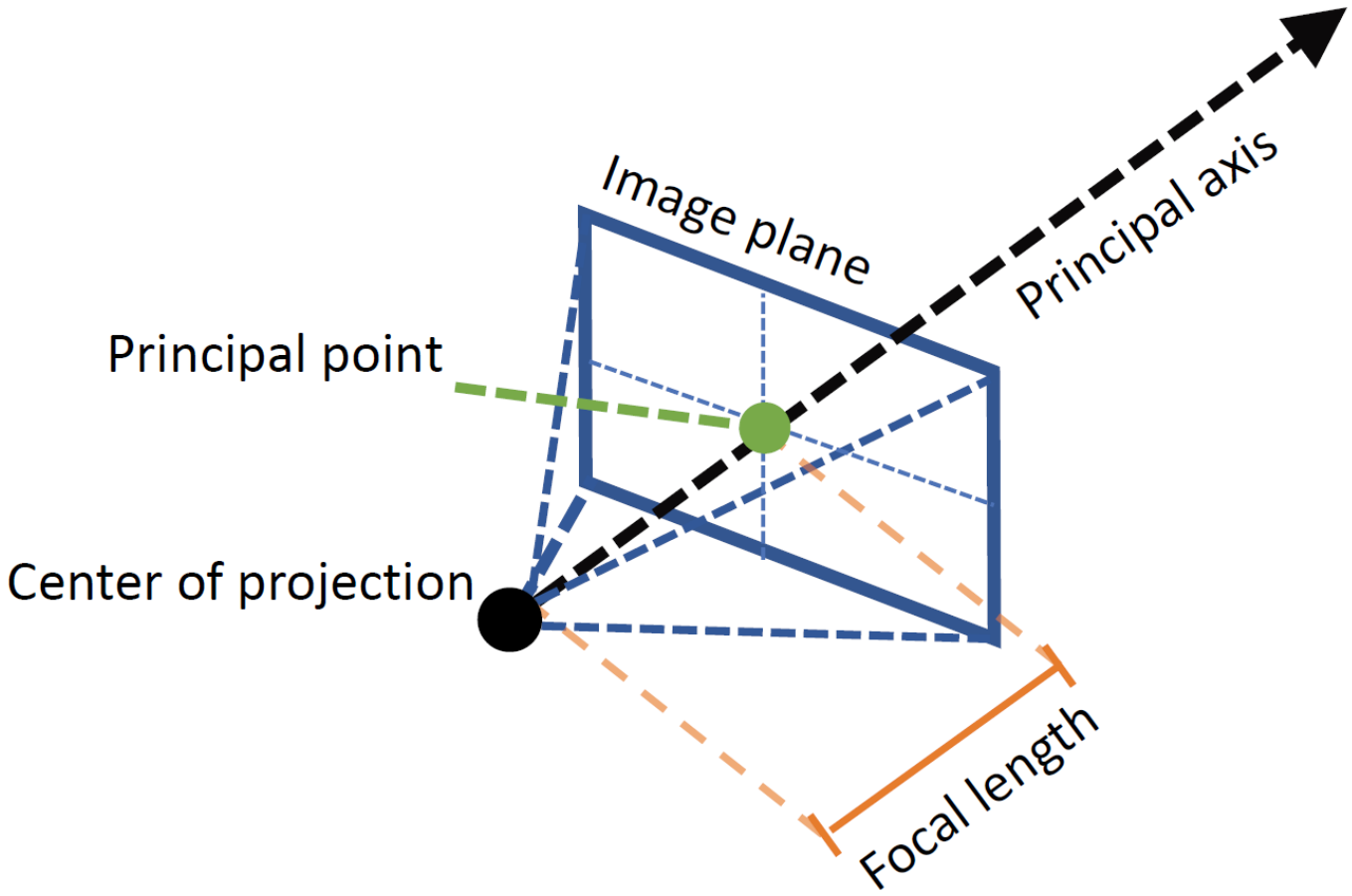
# Pinhole camera




Add a barrier to block off most of the rays

- This reduces blurring
- The opening known as the **aperture**
- How does this transform the image?

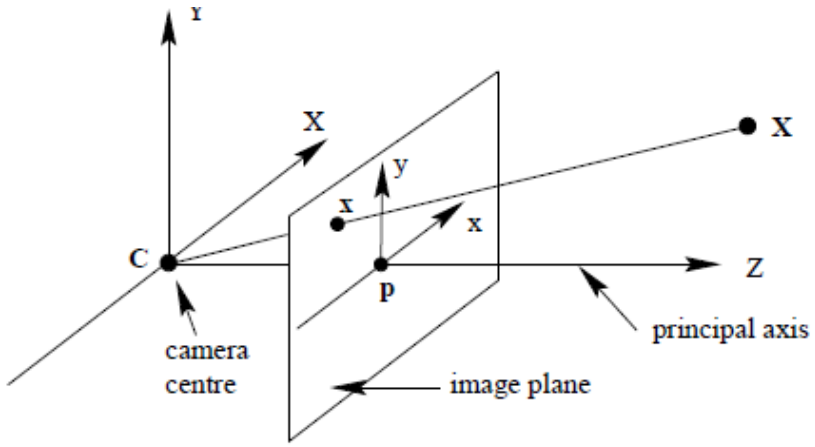
# Geometric Model: A Pinhole Camera

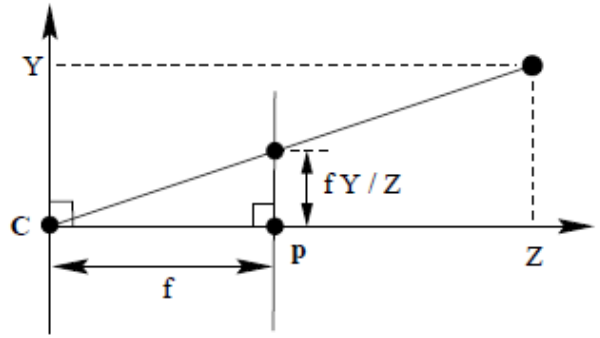


# Camera Models

- Starting with a model for the basic pinhole camera, we will develop models for the following cameras:
  1. CCD Imager Pinhole Camera,
  2. Finite Projective Camera,
  3. General Projective Camera,
  4. Orthographic Projective Camera,
  5. Scaled Orthographic Projective Camera, and
  6. Weak Perspective Projective Camera
- All of the above camera models are obtained by placing different kinds of constraints on the structure and the rank of a $3 \times 4$ matrix known as the Camera Projection Matrix

# The Basic Pinhole Model - Notations



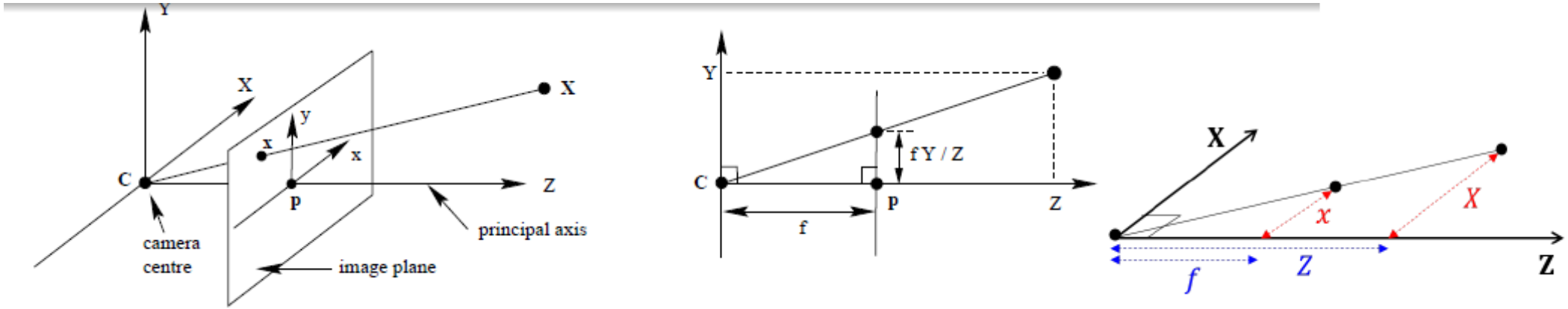




- p : Principle Point
- x: Image point (true image pixel)
- C: Center of Projection (COP) or optical center of the camera
- X: Object point $(X, Y, Z)^T$
- xy−coordinate system in the image plane: Center of the coordinate system of the image plane
  - Measurements usually done with respect to one of the corners not to the center of the sensor plane as shown in the figure above
- f : Focal length of the camera

# The Basic Pinhole Model

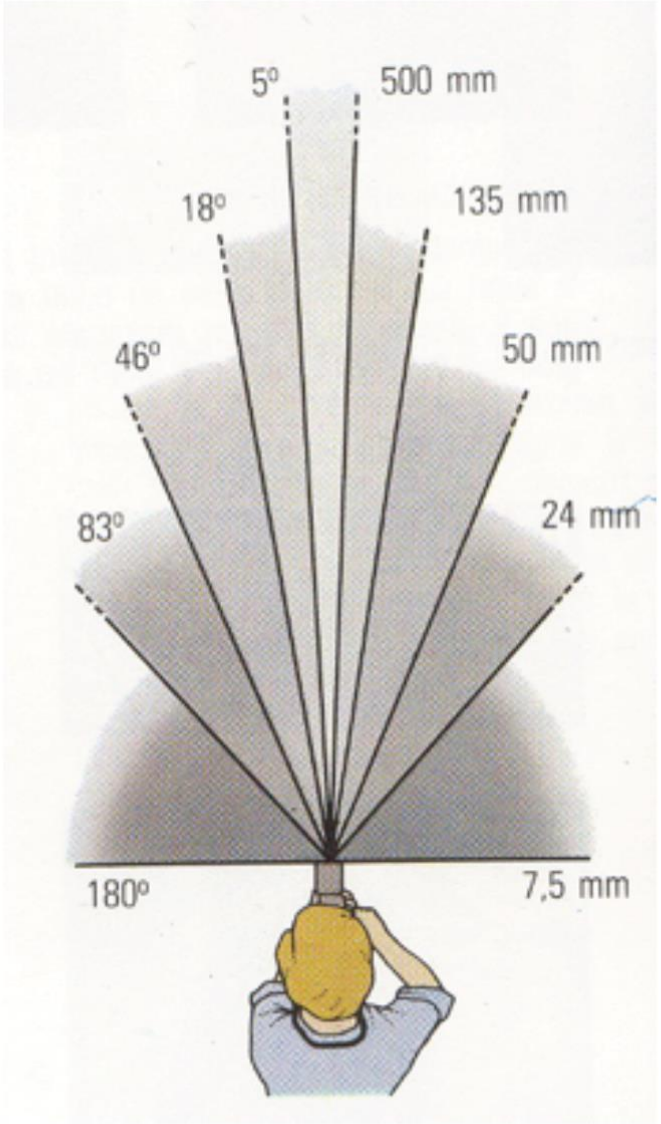
- Comparing similar triangles, the relationship between the world point $(X, Y, Z)^T$ and the corresponding image point $(x, y)$ is given by:
  - $x = fX/Z$, $y = fY/Z$, $z = Z$ (Nonlinear relationships)
- The above relationship is based on measuring the pixel coordinates $(x, y)$ with respect to **the principle point**.
  - More commonly, though, the pixel coordinates are measured with respect to a corner of the image frame.

# The Basic Pinhole Model




- $(X,Y,Z)^T \longmapsto (fX/Z, fY/Z) = (x,y)$
  Object Point  Image Point
  - This follows from Triangle Similarity
- $x = fX/Z$ or $(xZ = fX)$, $y = fY/Z$ or $(yZ = fY)$, $z = Z$ (Nonlinear relationships)
- Using homogeneous representation of the image and object (or world) points, we can write the following linear relationship between the pixel coordinates on the left and the corresponding world point coordinates on the right:
  - $$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} xz \\ yz \\ z \end{pmatrix} = \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$
    *HC Image Point*  *HC Object Point*

# Focal length in practice




24mm



50mm



135mm



# The Basic Pinhole Model

- $$\underbrace{\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}}_{\text{HC Image Point}} = \begin{pmatrix} xz \\ yz \\ z \end{pmatrix} = \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\text{HC Object Point}}$$
- $$\underbrace{\begin{pmatrix} xz \\ yz \\ z \end{pmatrix}}_{\text{HC Image Point } \vec{x}} = \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \underbrace{\begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{K} \left[ \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{I} \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}_{\vec{0}} \right] \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\text{HC Object Point } \vec{X}}$$
- $\vec{x} = K \left[ I | \vec{0} \right] \vec{X}$

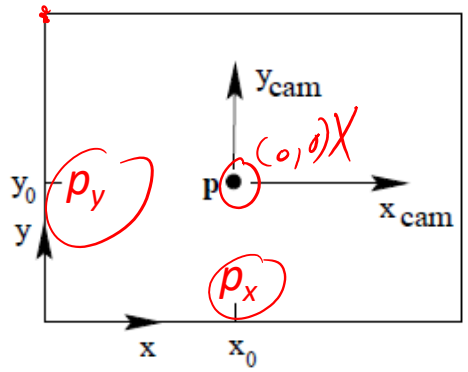
# The Basic Pinhole Model

- $\vec{x} = K\left[I|\vec{0}\right]\vec{X}$
- $K = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}$ : Intrinsic parameters of the camera
  - $\left[I|\vec{0}\right]$: Augmented Matrix
- $\vec{x}_{3\times1} = P_{3\times4}\vec{X}_{4\times1}$
  - $P_{3\times4} = K\left[I|\vec{0}\right]$ is the camera matrix

# The Basic Pinhole Model

- The world frame shown in the previous slides is in reality The Camera Coordinate Frame - because it is anchored at the camera.
- Therefore, what we have derived should be shown as: $\vec{x} = K \left[ I_{3\times 3} | \vec{0} \right] \vec{X_{cam}}$, where $\vec{X}_{cam}$ are the Camera Frame Coordinates of world $\vec{X}$

# The Basic Pinhole Model




- As we noted before, in practice, the image coordinates are always measured with respect to one of the sensor corner.
- Thus, let's go back to the previous basic pinhole model and change the coordinate frame of the image plane from the center to the left lower corner.
- Therefore, the principle point (PP) of the camera will be given with respect to the new origin of the coordinate frame of the image plane by $(p_x, p_y)$
- $(X, Y, Z)^T \longmapsto (fX/Z + p_x, fY/Z + p_y) = (x, y)$
  Object Point — Image Point

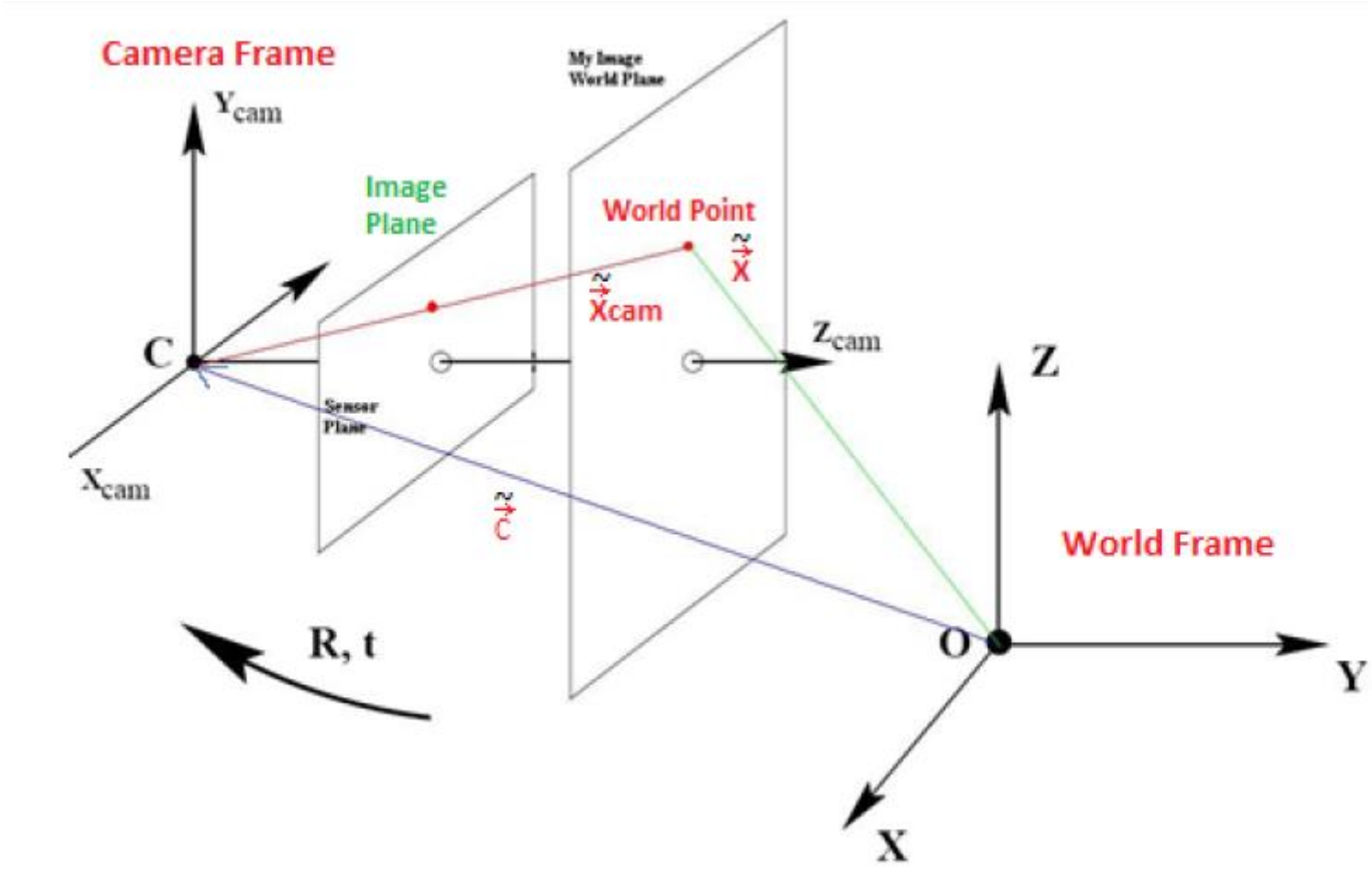
# The Basic Pinhole Model

- $$\underbrace{\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}}_{\text{HC Image Point}} = \begin{pmatrix} xz \\ yz \\ z \end{pmatrix} = \begin{pmatrix} fX + p_x Z \\ fY + p_y Z \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\text{HC Object Poi}}$$
  - $x = (fX + p_x Z)/Z$
  - $y = (fY + p_y Z)/Z$
- $\vec{x} = K\left[I|\vec{0}\right]\vec{X}$ or $\vec{x} = K\left[I|\vec{0}\right]\vec{X_{cam}}$
- $K = \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix}$: Intrinsic parameters of the camera (or the Camera Calibration Matrix $P_{3\times 4}$)

# Relationships between the camera coordinate frame and the world frame

- Let's now separate the camera coordinate frame and the world frame
- In order to find the relationship between $\vec{X}_{cam}$ and $\vec{X}$, in camera modeling work, it is often best to first establish the relationship between physical coordinate that we denote by $\tilde{\vec{X}}_{cam}$ and $\tilde{\vec{X}}$

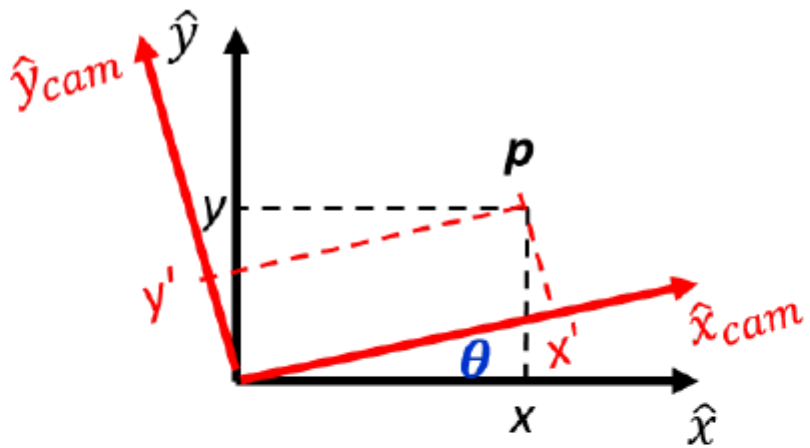
# Relationships between the camera coordinate frame and the world frame



# Relationships between the camera coordinate frame and the world frame

- Let $R$ denote the rotation from the physical world coordinates to the camera frame coordinates.
  - To understand that what we mean by 'from' and 'to' clauses, assume that the two origins are at exactly the same point.
  - Our definition of $R$ would imply that $\tilde{\vec{X}}_{cam} = R\tilde{\vec{X}}$.
  - To see this more clearly, consider the 2D case at the next slide:

# Relationships between the camera coordinate frame and the world frame




- $\tilde{\vec{X}}_{cam} = \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = R\tilde{\vec{X}}.$
- So, when $R$ multiplies the coordinates of a point in the world frame, we get the coordinates of the **same point** in the camera frame.

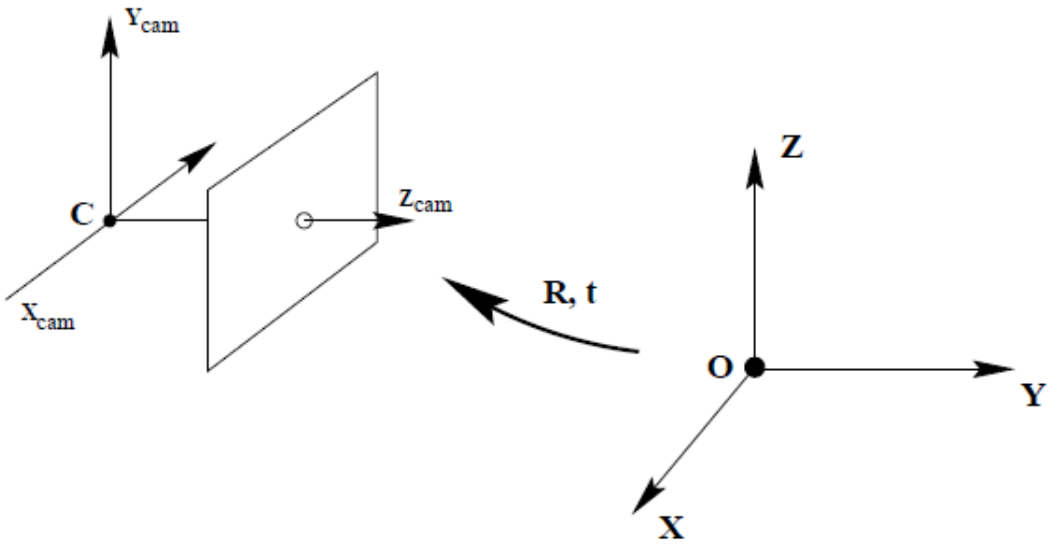
# Extrinsic Parameters of the Camera

- Let's now denote the relationships between the camera coordinate frame and the world frame by the extrinsic parameters of the camera.

# Extrinsic Parameters of the Camera

- Let $\tilde{C}$ denote the transnational vector from the world frame origin to the camera frame origin.
- If we ignore the rotational misalignment between the two frames, we have $\tilde{\vec{X}}_{cam} = \tilde{\vec{X}} - \tilde{\vec{C}}$.
  - When we have both a translational $\tilde{\vec{C}}$ and a rotation $R$ from the world frame to the camera frame, we have $\tilde{\vec{X}}_{cam} = R\left(\tilde{\vec{X}} - \tilde{\vec{C}}\right)$.
    - For the world point $\tilde{\vec{X}}$, the subtraction $\left(\tilde{\vec{X}} - \tilde{\vec{C}}\right)$ gives us the coordinates of the same world point in the camera frame at a vector distance $\tilde{\vec{C}}$ but with the camera frame still aligned with the world frame. That by multiplying this by $R$ should yield $\tilde{\vec{X}}_{cam}$ follows from our previous explanation of how $R$ works.

# Extrinsic Parameters of the Camera




- Camera rotation and translation with respect to the world frame that is shown to the right of the figure above.
- $\widetilde{X}_{cam} = R(\widetilde{X} - \widetilde{C})$
  - $\widetilde{X}_{cam}$ and $\widetilde{X}$ are in physical coordinates
  - Do translation first and then do the rotation part

# Extrinsic Parameters of the Camera

- The equation $\tilde{\vec{X}}_{cam} = R\left(\tilde{\vec{X}} - \tilde{\vec{C}}\right)$ can be adapted to the homogeneous coordinates by:
- $$\vec{X}_{cam} = \begin{pmatrix} \tilde{\vec{X}}_{cam} \\ 1 \end{pmatrix} = \begin{bmatrix} R & -R\tilde{\vec{C}} \\ 0 & 1 \end{bmatrix} \begin{pmatrix} \tilde{\vec{X}} \\ 1 \end{pmatrix} = \begin{bmatrix} R & -R\tilde{\vec{C}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} =$$
  $$\begin{bmatrix} R & -R\tilde{\vec{C}} \\ 0 & 1 \end{bmatrix} \vec{X}$$
- Substituting the relationship between the homogeneous vectors $\vec{X}_{cam}$ and $\vec{X}$ in the pinhole camera model shown before, we can write for the pixel coordinates:
  - $\vec{x} = K\,[I|\vec{0}]\,\vec{X}_{cam}$
  - $\vec{x} = K\,[I|\vec{0}] \begin{bmatrix} R & -R\tilde{\vec{C}} \\ \vec{0}^T & 1 \end{bmatrix} \vec{X}$

# Extrinsic Parameters of the Camera

- $\vec{x} = K\,[I|\vec{0}] \begin{bmatrix} R & -R\tilde{\vec{C}} \\ \vec{0}^T & 1 \end{bmatrix} \vec{X}$
- Note: $\left[I_{3\times 3}|\vec{0}\right] \begin{bmatrix} R & -R\tilde{\vec{C}} \\ \vec{0}^T & 1 \end{bmatrix} =$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & (-R\tilde{\vec{C}})_x \\ r_{21} & r_{22} & r_{23} & (-R\tilde{\vec{C}})_y \\ r_{31} & r_{32} & r_{33} & (-R\tilde{\vec{C}})_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & (-R\tilde{\vec{C}})_x \\ r_{21} & r_{22} & r_{23} & (-R\tilde{\vec{C}})_y \\ r_{31} & r_{32} & r_{33} & (-R\tilde{\vec{C}})_z \end{bmatrix}$$

- $= \left[R|-R\tilde{\vec{C}}\right] = R\left[I|-\tilde{\vec{C}}\right]$

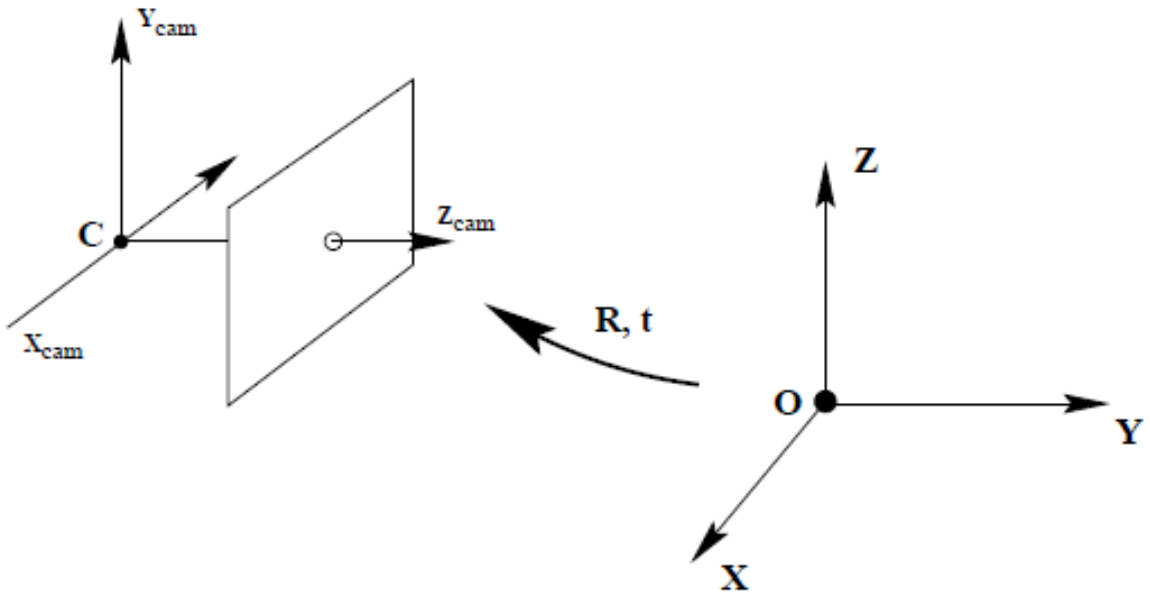
# Extrinsic Parameters of the Camera

- Thus, $\vec{x} = K\left[R|-R\tilde{\vec{C}}\right]\vec{X} = KR\left[I|-\tilde{\vec{C}}\right]\vec{X} = \boldsymbol{K[R|t]}\vec{X} = \boldsymbol{P}\vec{X}$

  - $\vec{x}$: pixel coords (homogeneous)
  - $K$: camera parameters (intrinsic)
  - $R$: World-to-Camera rotation (physical)
  - $\tilde{\vec{C}}$: World-to-Camera Translation (physical)
  - $\vec{t} = -R\tilde{\vec{C}}$
  - $\vec{X}$: World point (homogeneous)

**The equation above will serve as our fundamental equation for modeling a pinhole camera.**

# Extrinsic Parameters of the Camera



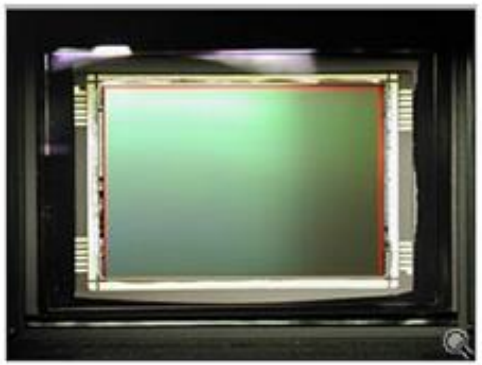

- $\vec{x} = K\,[I|\vec{0}] \begin{bmatrix} R & -R\tilde{C} \\ 0 & 1 \end{bmatrix} \vec{X} = K\left[R|-R\tilde{C}\right]\vec{X} = KR\left[I|-\tilde{C}\right]\vec{X} = P\vec{X}$
  - $P = KR\left[I|-\tilde{C}\right] = K\left[R|-R\tilde{C}\right] = K\,[R|t]$
    - $t = -R\tilde{C}$

# Taking into account the sampling of the image plane

- For images recorded with digital cameras (e.g. CCD cameras), we must also factor in the sampling rates in the image plane.
- Let's start by assuming the that the 2D array of photo sensors in the image plane is perfectly rectangular, with $m_x$ cells per unit length along $x$ and $m_y$ cells per unit length along $y$.

# Taking into account the sampling of the image plane



- So recall that we assumed the CCD image sensor have $m_x$ elements per unit length along $x$ and $m_y$ elements per unit length along $y$.
- We may now associate the "**discretized**" location $(I, J)$ with the pixel at $(x, y)$, with:
  - $I = \boldsymbol{m_x} x$ and $J = \boldsymbol{m_y} y$
    - Note 1: $I$ and $J$ are known as the pixel coordinates in image **(True pixels)**
    - Note 2: Obviously, the products $m_x x$ and $m_y y$ do not by themselves yield integer values . However, to get integers indices associated with the cells, all we have to do is to ignore the fractional parts of $m_x x$ and $m_y y$ .

# Taking into account the sampling of the image plane

- Using our foundational equations for $x$ and $y$ that we have discussed before in the the Basic Pinhole Model, we can write:
  - $I = m_x x = m_x(fX/Z + p_x) = m_x fX/Z + \boldsymbol{m_x p_x}$
  - $J = m_y y = m_x(fY/Z + p_y) = m_y fY/Z + \boldsymbol{m_y p_y}$
- If we use $\boldsymbol{x_0 = m_x p_x}$ and $\boldsymbol{y_0 = m_y p_y}$ as the integer coordinates of the principle point in the image plane, we can write:
  - $I = m_x fX/Z + \boldsymbol{x_0}$ and $J = m_y fY/Z + \boldsymbol{y_0}$
  - $I = \frac{m_x fX + x_0 Z}{Z}$ or $\boldsymbol{IZ = m_x fX + x_0 Z}$, and $J = \frac{m_y fY + y_0 Z}{Z}$ or $\boldsymbol{JZ = m_y fY + y_0 Z}$
  - These last two equations can be turned into the following composite relationship between the homogeneous representations of the "discrete" pixel coordinates and the world point in the camera coordinate frame as shown in the next slide.

# Taking into account the sampling of the image plane

- $$\begin{pmatrix} I \\ J \\ 1 \end{pmatrix} = \begin{pmatrix} IZ \\ JZ \\ Z \end{pmatrix} = \begin{pmatrix} m_x f X + x_0 Z \\ m_y f Y + y_0 Z \\ Z \end{pmatrix} =$$

$$\begin{bmatrix} m_x f & 0 & x_0 & 0 \\ 0 & m_y f & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

  - with $\alpha_x = m_x f$, $\alpha_y = m_y f$, and $(x_0, y_0)$: the principle point is in image plane
  - You can think of:
    - $\alpha_x$ as the focal length in terms of the number of pixels using the $x$-sampling rate (or $x$ digitization), and of
    - $\alpha_y$ the focal length in terms of the number of pixels using the $y$-sampling rate (of $y$ digitization).

# Taking into account the sampling of the image plane

- $\begin{pmatrix} IZ \\ JZ \\ Z \end{pmatrix}$ is obviously the homogeneous representation of the discrete pixel coordinates $(I, J)$.
  - Going forward, we will represent this homogeneous vector by $\vec{x}$.
- So we again have:
  - $\vec{x} = K \left[ I_{3\times 3} | \vec{0} \right] \vec{X}_{cam}$, with $K = \begin{bmatrix} m_x f & 0 & x_0 \\ 0 & m_y f & y_0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$
  - This relationship between the homogeneous vectors $\vec{x}$ and $\vec{X}_{cam}$ is the same as what we saw before. **The only difference here is the form of the intrinsic parameters of the camera calibration matrix $K$.**

# Taking into account the sampling of the image plane

- Therefore, our overall relationship between a world point $\vec{X}$ and the corresponding pixel $\vec{x}$ remains the same as shown earlier except for the intrinsic calibration matrix $K$ being as shown in the previous slide.

- Thus, we produce below one of those two relationship:
  - $\vec{x} = \underbrace{KR\left[I_{3\times3}|-\tilde{\vec{C}}\right]}_{\text{Projection Matrix P}}\vec{X} = P_{3\times4}\vec{X}$
  - Note: $P_{3\times4}$ is homogeneous and $K = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$

# Camera Matrix: Degree of Freedoms

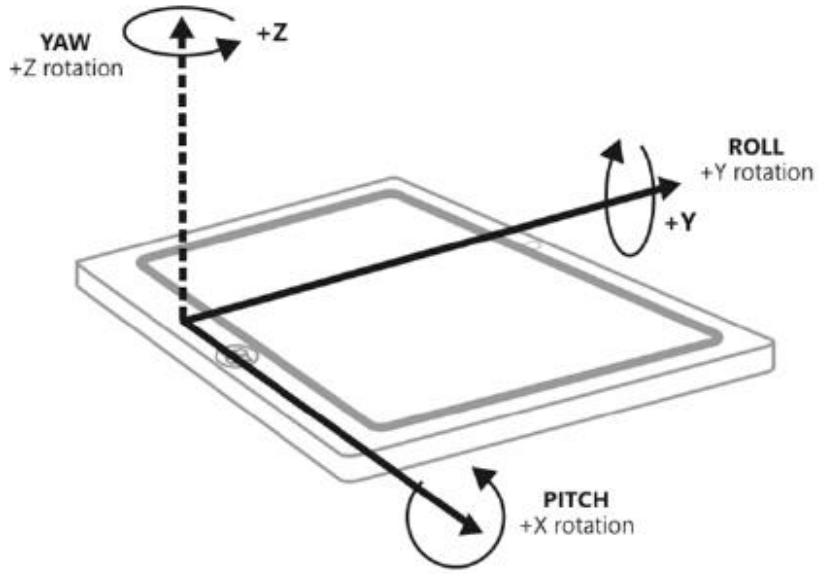
- $P_{3\times4} = KR\left[I_{3\times3}|-\tilde{\vec{C}}\right] = K\left[R|-R\tilde{\vec{C}}\right] = K_{3\times3}\left[R_{3\times3}|\vec{t}_{3\times1}\right]$
- **The $3\times4$ projection matrix $P$ has 10 Degrees of Freedom (DoF).**
  - Of these 10 DoF: $K$ accounts for 4, $R$ for 3, and $\tilde{\vec{C}}$ for the remaining 3.
  - $K = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3}$, 4 DoF
  - $R_{3\times3}$, Euler angles: yaw, pitch, and roll. 3 DoF
  - $t_{3\times1} = -R\tilde{\vec{C}}$, 3 DoF
  - Total: **4 DoF + 3 DoF + 3 DoF = 10 DoF**
- But $P_{3\times4} = 12$ parameters, thus **we should have 11 DoF.** Therefore **we are missing 1 DoF**.

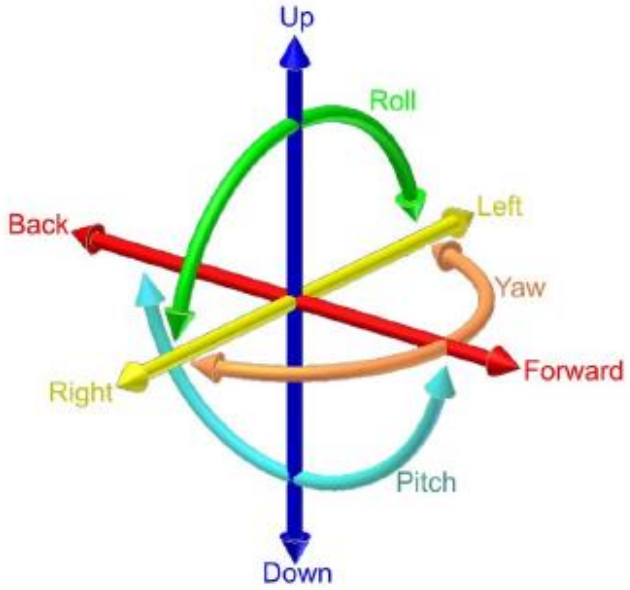
# Aside: Euler Angles

- $R_{3\times3} = R_x(\gamma)R_y(\beta)R_z(\alpha) =$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\gamma & -sin\gamma \\ 0 & sin\gamma & cos\gamma \end{bmatrix} \begin{bmatrix} cos\beta & 0 & sin\beta \\ 0 & 1 & 0 \\ -sin\beta & 0 & cos\beta \end{bmatrix} \begin{bmatrix} cos\alpha & -sin\alpha & 0 \\ sin\alpha & cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

  - Euler angles: yaw, pitch, and roll.



# Camera Matrix: Degree of Freedoms

- Now consider that since the $3 \times 4$ $P$ has a maximum of 11 DoF, as to what might be the practical significance of the remaining single DoF in $P$.
- Consider the following form for $K$:

# Camera Matrix: Degree of Freedoms

- Missing DoF
- $K = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$
  - 5 DoF
  - $s$ : skew factor as a result of skew coordinate (rectangular representation). Will be called the skew parameter.
- With this $K$, the projection matrix $P$ will have the maximum allowable DoF of 11.
  - **As it turns out, the additional parameter $s$ in $K$ is required if the image sensor array is not perfectly rectangular.**
  - In digital cameras, there is always a bit of "skew" in the $xy$-layout of the sensor cells.

# General Notes

- $K$ matrix is non-singular and will never be singular
- $R$ matrix is by definition non-singular or it won't be a rotation matrix
- $\vec{t}$ vector can go to zero (no translation)

# Recall: The Pinhole Camera Model

- $P = KR\left[I|-\tilde{C}\right] = K\left[R|-R\tilde{C}\right] = K[R|t]$
  - $K = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$
  - $s = 0$: Ideal Camera

# Finite Projective Cameras

- In general, and specially if one allows for virtual cameras, the relationship between a world point $\vec{X}$ and its corresponding pixel at $\vec{x}$ is expressed as:
- $\vec{x}_{3\times1} = \boldsymbol{P}_{3\times4}\vec{X}_{4\times1}$
  - In general , we will express $\boldsymbol{P}$ as $P = \begin{bmatrix} \vec{p}_1 & \vec{p}_2 & \vec{p}_3 & \vec{p}_4 \end{bmatrix}$ when we need to identify its four columns separately.
  - And, sometimes, we will show $\boldsymbol{P}$ as $P = [M|\vec{p}_4]$, where $M$ is the $3\times3$ submatrix formed by the first three columns of $P$.
- **We now claim that any $3\times4$ real matrix whose first three columns constitute a non-singular submatrix is a valid camera.**
  - We refer to the set of all such cameras as **Finite Projective Cameras**.

# Notes on the Finite Projective Cameras

- $P = KR\left[I|-\widetilde{C}\right] = M\left[I_{3\times 3}|M^{-1}\vec{p}_4\right] = [M|\vec{p}_4]$
  - $M = KR$, and $\widetilde{C} = -M^{-1}\vec{p}_4$
- For pinhole and finite projective cameras the left $3 \times 3$ sub-matrix of $P$ is non-singular.

# Finite Projective Cameras: Summary

- Obviously, all digital cameras are examples of Finite Projective Cameras. However, the **Finite Projective Set** includes cameras that cannot be realized physically but that may nonetheless be useful as **virtual cameras**.
- When a Finite Projective Camera cannot be realized physically, it may be:
  - on account of where the COP is in relation to the object, or
  - on account of where the image is formed on the image plane, or
  - on account of the values given to the intrinsic parameters of the camera.

# General Projective Camera

- The Finite Projective Camera is obviously a more general category than the pinhole camera (with and without the skew parameter).
- An even more general category of cameras is **the General Projective Camera**.
  - **The only constraint we now place on the projection matrix $P$ is that its rank be exactly 3.**

# Camera at Infinity

- Despite the sense conveyed by "infinity" in the title of this slide, these cameras are of great practical importance when images are formed under orthographic and weak perspective conditions.
- Recall:
  - **Any $3 \times 4$ matrix is a projective camera provided its rank is 3.**
  - A general projective camera (that is NOT finite projective) has the following property:
    - The submatrx $M$ of $P = [M|\vec{p}_4]$ is singular and that the null vector $\vec{d}$ of $M$ gives us the camera center $\vec{C} = \begin{pmatrix} \vec{d} \\ 0 \end{pmatrix}$ at infinity for such a camera.

# Camera at Infinity

- A general projective camera located at infinity.
  - Has parallel beams of rays parallel to either $x$-direction (world $X$), or $y$-direction (world $Y$) or $z$-direction (world $Z$).
- We will now consider a sub-category of such general projective cameras - the sub-category (or Special Case) is known as the Affine Camera

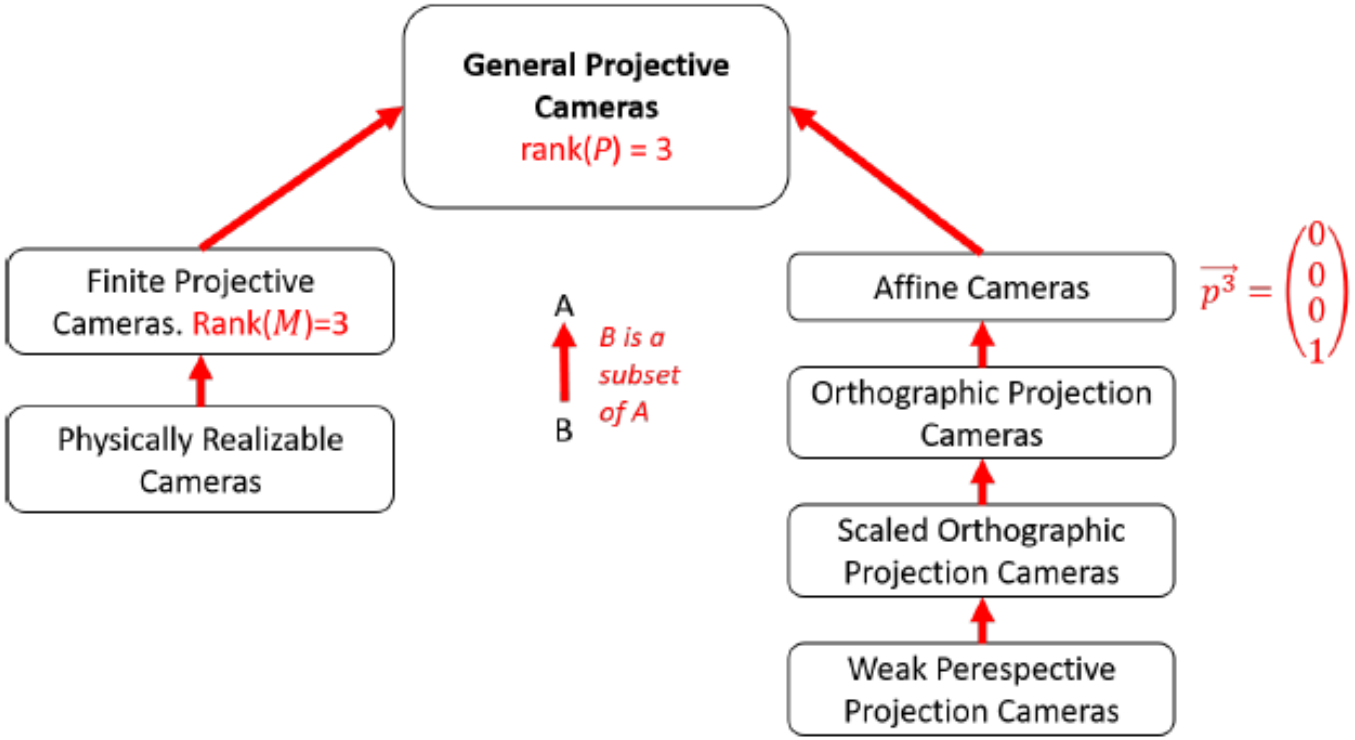
# Affine Cameras

- Under Affine Cameras, we will consider:
  - Orthographic Projection Camera (rays are parallel),
  - Scaled Orthographic Projection Camera, and
  - Weak Perspective Projection Camera
- In general, **a camera is affine if the last row of the $3 \times 4$ $P$ is** $\begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}$.
  - This means the principle plane $\pi_\infty$ which is the same as plane at infinity is given by $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}^T$
  - Affine camera maps parallel lines to parallel lines
  - Obviously, with $P = [M|\vec{p}_4]$, $M$ for an affine camera is given by

$$M = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ 0 & 0 & 0 \end{bmatrix}$$

# Important Note: Moving Forward

- You will see that all Central Projection Cameras of practical interest can be mapped to the following hierarchy of cameras:




***Note:*** *All of these different types of cameras can be used for Computational Photography*

# Orthographic Projection Camera

- In its generic form, this affine camera (i.e. the Orthographic camera which is a special case of an affine camera.) is defined by:
  - $P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, $M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $\vec{p}_4 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$
  - Since $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ is the null vector of $M$ in this case, the CoP of the camera is at $\begin{bmatrix} \vec{d} \\ 0 \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$, which is a point at $\infty$ along the world $Z$-axis, this camera gives us an orthographic projection of the object onto the $xy$-plane.
  - By orthographic, we mean projection with **a bundle of parallel rays**.
    - Obviously, only those object points would be projected that are visible to the camera.

# Orthographic Projection Camera

- For other poses of the camera, we introduce a rotation $R$ and a translation $\vec{t}$ from the world frame to the camera frame.
- Following the same logic as we did before at the beginning of these slides, we can write:
  - $\vec{x} = P\vec{X} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R & \vec{t} \\ \vec{0}^T & 1 \end{bmatrix} \vec{X} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \vec{r}_1^T & t_1 \\ \vec{r}_2^T & t_2 \\ \vec{r}_3^T & t_3 \\ \vec{0}^T & 1 \end{bmatrix} \vec{X}$, where $\vec{r}_1^T$, $\vec{r}_2^T$, and $\vec{r}_3^T$ are the three row vectors of $R$, and where $t_1$, $t_2$, $t_3$ are the three components of $\vec{t}$.
  - Therefore, the projection matrix $P$ of an orthoprojection camera at pose $(R, \vec{t})$ with respect to the world frame is given by $P = \begin{bmatrix} \vec{r}_1^T & t_1 \\ \vec{r}_2^T & t_2 \\ \vec{0}^T & 1 \end{bmatrix}$.
    - By World frame here, we mean the coordinate frame in which the generic configuration is defined.

# Scaled Orthographic Projection Camera

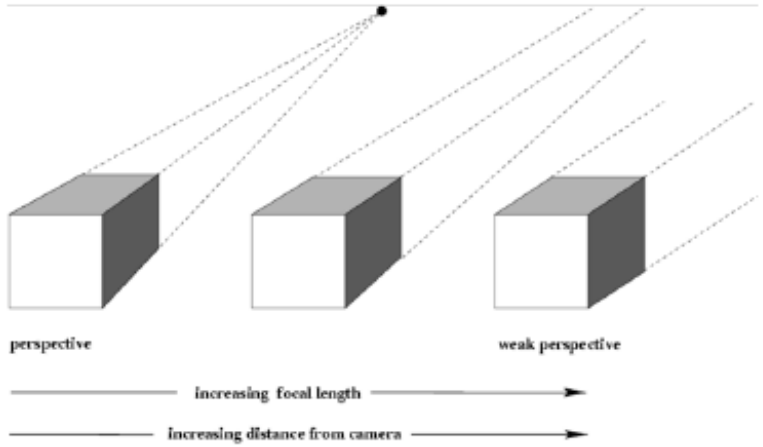
- A Scaled Orthographic Projection Camera is a regular Orthographic Projection Camera as described before that is followed by isotropic scaling of the image.
  - This means that whatever image is produced by the $P$ matrix of the previous slide needs to be scaled by a factor $k$.
  - Therefore, the new $P$ would be given by:
    - $$P = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \vec{r}_1^T & t_1 \\ \vec{r}_2^T & t_2 \\ \vec{0}^T & 1 \end{bmatrix} = \begin{bmatrix} \vec{r}_1^T & t_1 \\ \vec{r}_2^T & t_2 \\ \vec{0}^T & 1/k \end{bmatrix}.$$
    - Note: for both the $P$ here and in the previous case, the first two rows of $M$ submatrix of $P$ must be orthogonal.

# Weak Perspective Projection Camera

- This camera differs from the one in the previous case in only one sense:
  - We now use two different scale factors for the $x$ and the $y$ directions in the image plane.
  - Consequently, we can write the projection matrix as:
  - $$P = \begin{bmatrix} \alpha_x & 0 & 0 \\ 0 & \alpha_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \vec{r}_1^T & t_1 \\ \vec{r}_2^T & t_2 \\ \vec{0}^T & 1 \end{bmatrix}.$$
    - Note: The first two rows of $M$ submatrix of $P$ would still be orthogonal.

# Important Notes

- The last two affine camera models (Scaled and Weak-Perspective Cameras) are often used for images of distant objects recorded with long focal length lenses and when the objects subtend small angles at the camera center.








# Imaging of Various Geometrical Entities: Action of Projective Camera on Point

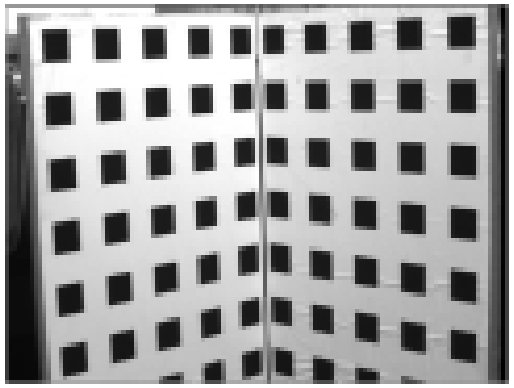
- Forward projection
- $\vec{x}_{1\times 3} = P_{3\times 4}\vec{X}_{1\times 4} = [M|\vec{p}_4]\,\vec{X}_{1\times 4} = \begin{bmatrix} \vec{p}_1 & \vec{p}_2 & \vec{p}_3 & \vec{p}_4 \end{bmatrix}\vec{X}_{1\times 4}$
  - Back-projection
    - $\vec{X} = P^{\dagger}\vec{x}$
    - $P^{\dagger}$:pseudo-inverse $P^{\dagger} = P^T(PP^T)^{-1}$
    - $PP^{\dagger} = I$

# Imaging of Various Geometrical Entities

| Geometrical Entity | Forward Projection | Backward Projection |
|---|---|---|
| **Points** | 3D world points $\rightarrow$ 2D image points | 2D image point $\rightarrow$ 3D line |
| **Lines** | 3D world line $\rightarrow$ 2D image line | 2D image line $\rightarrow$3D plane |
| **Planes** | 3D world plane $\rightarrow$ homography | Does not make sense to apply |
| **Conics** | Does not make sense to apply | 2D conic $\rightarrow$ cone |
| **Quadrics** | Quadric $\rightarrow$ conic | Does not make sense to apply |
| **Absolute Conic** | Image of absolute conic $\Omega_\infty$ is $K^{-T}K^{-1}$ | Does not make sense to apply |

- The fact that the Image of absolute conic $\Omega_\infty$ is $K^{-T}K^{-1}$ is used in camera calibration.

# Camera Calibration (Will cover it if time allows)



- Main goal is to compute the intrinsic parameters $K$ of the camera and the extrinsic parameters $R \& \vec{t}$
  - Computer must know the intrinsic parameters $K$ of the camera and the extrinsic parameters $R \& \vec{t}$ as fast as possible (instantly) in many applications.

# Questions

The End